

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : {FACULTYNAME}

Semester : SEMESTER - 6

Subject Code & Name: BTEEC601 - SWITCHGEAR AND PROTECTION

Time : 3 hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

In a power system, backup protection is provided to ensure:

- A) Primary relay fails to trip B) Circuit breaker fails to operate C) CT or PT fails D) Overload protection is not possible

The zone of protection is primarily defined by:

- A) Power flow direction B) The type of relay used C) The location of CTs and PTs D) Type of fault expected

3 A MHO relay is preferred over an impedance relay for distance protection because:

- A) It is not affected by arc resistance B) It is inherently directional C) It operates on zero sequence current D) It is sensitive to voltage dips

4 The key difference between amplitude and phase comparators in static relays lies in:

- A) Sensitivity to DC offset B) Frequency range of operation C) Response to phase angle difference D) Ability to operate without CTs

In numerical relays, sampling frequency affects:

- A) Relay burden B) Accuracy and speed C) Number of zones D) Trip circuit configuration

6 The rate of rise of restriking voltage (RRRV) in a circuit breaker depends on:

- A) Type of arc extinction medium B) Frequency and power factor of the system C) Closing time of CB

- 7 Which statement about SF₆ circuit breakers is incorrect?
- A) SF₆ has excellent dielectric properties B) It supports self-arc extinction C) It is used in low voltage applications D) It is non-toxic and inert under normal conditions

- 8 Current chopping is undesirable in CB operation because it:
- A) Delays arc extinction B) Causes over-voltage transients C) Increases contact erosion D) Prevents successful tripping

In carrier-aided protection, the main advantage is:

- A) Simpler wiring B) Reduced circuit breaker ratings C) Faster fault isolation across zones D) Use of low-cost relays

Instantaneous overcurrent relays are not suitable for long transmission lines because:

- A) They are not directional B) They cannot handle high current C) They operate on differential principles D) Fault currents may be too low

- 11 In transformer differential protection, inrush current is prevented from causing false trips by:

- A) Using higher tap settings B) Employing harmonic restraint C) Delaying the tripping time D) Using backup relays

Which of the following is not a cause of unbalanced loading in alternators?

- A) Open phase conductor B) Unequal phase voltages C) Faulty stator winding D) Reverse power flow

Q2. Solve the following.

- A) Discuss the philosophy of protection in a power system. Explain the concepts of primary and backup protection and zones of protection with suitable diagrams. 6
- B) Describe the working principles of different types of relays: attracted armature type, balanced beam type, and induction type. Illustrate their construction and applications. 6

Q3. Solve the following.

- A) Compare and contrast amplitude comparator and phase comparator techniques. How are these techniques implemented in static relays? 6
- B) Explain working principle, relaying of numerical relays. 6

Q4. Solve Any Two of the following.

- A) Describe the phenomenon of arcing in circuit breakers. Explain the arc interruption process in detail. 6
- B) Explain the working principle, construction, advantages, and disadvantages of following circuit breakers: 6
- A) SF₆ circuit breaker.
- B) Vacuum circuit breaker.
- C) Describe with neat sketch, time graded and current graded protection of a feeder. 6

Q5. Solve Any Two of the following.

- A) Two relays R1 and R2 of 5A rating are connected in two sections to a feeder CT's are of ratio 1000/5 A. The PSM of relay R1 is 100% and R2 is 125%. The operating time characteristics of the relays is given below: 6

PSM	2	4	5	8	10	20
Operating Time(Sec)	10	5	4	3	2.8	2.4

The TSM of the relay R1 is 0.3. The time grading scheme has a discriminative time margin of 0.5sec. Between the relay. A three phase short circuit at F results in fault current of 5000A. Find the actual operating times of R1 and R2. What is TMS of R2.

- Q6. Solve Any Two of the following.
- Explain the working of overcurrent protection and directional overcurrent relays. Describe their applications in radial and ring main feeder systems. 6
 - Explain three step protection scheme for transmission line protection in detail. 6
 - Explain the different types of faults in alternators and describe the protection schemes used for stator and rotor protection. 6
 - Describe differential protection of transformers. What difficulties are encountered in transformer differential protection and how are they overcome? 6
 - Explain buchholz relay in detail? 6